ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

M.Tech. (CSE)

Semester: II

SUBJECT CODE: MCS204

OBJECT ORIENTED MODELING

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question.

Q. 1. Define the following terms: **(1×7 = 7 Marks)**

(i) UML
(ii) Use Case
(iii) OOAD
(iv) Class Diagram
(v) Design Pattern
(vi) OCL
(vii) MDA

Q. 2. Answer the following questions: **(1×7 = 7 Marks)**

(i) What are the advantages of UML?
(ii) Describe the elements of a use case diagram.
(iii) Explain the importance of object-oriented design.
(iv) What is a CASE tool?
(v) What do you understand by meta-modeling?
(vi) Why is round-trip engineering important?
(vii) Define model-based testing.

SECTION – B

NOTE:Attempt all questions. **(7×2×2 = 28 Marks)**

Q.3. (i) (a) Explain the different types of UML diagrams with examples.

OR

(b) How does use case modeling help in capturing requirements?

(ii)(a) Describe any two object-oriented design methodologies.

OR

(b) Compare functional modeling and object-oriented modeling.

Q.4. (i)(a) What are design patterns? Explain any two with diagrams.

OR

(b) Discuss the role and advantages of CASE tools in software Modeling.

(ii)(a) What is OCL? How is it used for precise modeling?

OR

(b) Analyze a sample software model and demonstrate how Constraints are applied using OCL.

SECTION – C

NOTE: Attempt all questions. (14×2 = 28 Marks)

Q. 5. (a) What is Model-Driven Architecture (MDA)? Explain its architecture and benefits.

OR

(b) Discuss the process and significance of meta-modeling in software engineering.

Q. 6. (a) Write short notes on any two:

i. Aspect-Oriented Modeling
ii. Modeling Non-functional Properties
iii. UML Profiles

OR

(b) What are open research issues in OOAD and modeling? Suggest any two potential solutions for future directions.